LLM-Guided SAM for Left Ventricle Segmentation and Functional Analysis in Echocardiography

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Background & Motivation

Clinical Context: LV segmentation is a fundamental step in echocardiographic functional assessment. Clinically important indices such as EI and ACR still depend on manual contouring, which is time-consuming, subjective, and difficult to standardize.

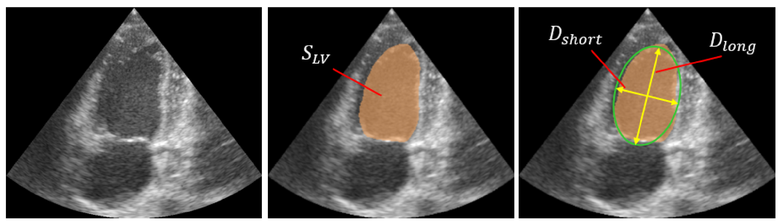


Figure 1. Clinical workflow for LV functional assessment in echocardiography: (a) selection of key frames; (b) annotation of the endocardial boundary to obtain S_{LV} for ACR; (c) ellipse fitting to derive D_{short} and D_{long} for EI.

Challenges:

- Speckle noise, low contrast, and blurred boundaries frequently obscure LV contours, making echocardiographic segmentation challenging under real clinical imaging conditions.
- Existing methods often lack robust anatomical prompting and cardiac-aware structural priors, resulting in poor generalization across patients, views, and imaging protocols.
- Functional validation is often overlooked, with segmentation outputs rarely linked to clinical indices such as EI and ACR.

Objective: To build an automatic LV segmentation framework that improves robustness in echocardiography and supports direct functional evaluation using EI and ACR.

Overview of LV-SAM++

LV-SAM++ inherits the SAM backbone but introduces three targeted modules: (1) a Local Feature Enhancement (LFE) branch, (2) a Semantic Prompt Learning (SPL) branch with Prompt-aware Feature Weighting (PaFW), and (3) a functional assessment module for deriving indices such as eccentricity index (EI) and area change ratio (ACR).

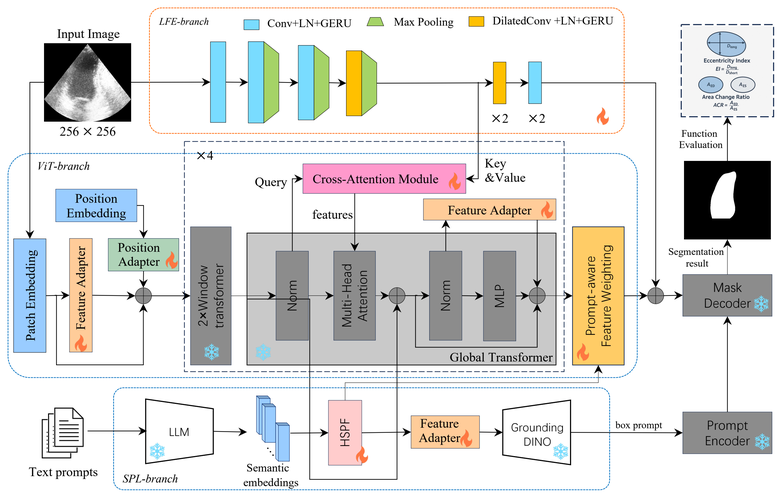


Figure 2. Overview of **LV-SAM++**: LFE branch enhances boundary details, SPL branch integrates semantic prompts with PaFW, and the functional module computes EI and ACR.

Functional Assessment and Loss Function

Functional Assessment

From predicted LV masks, two indices are computed:

$$EI = \sqrt{1 - \left(\frac{L_{\text{minor}}}{L_{\text{major}}}\right)^2}, \qquad ACR = \frac{A_{ED} - A_{ES}}{A_{ED}}. \tag{1}$$

where $L_{\text{major}}, L_{\text{minor}}$ are ellipse axes, and A_{ED}, A_{ES} denote LV areas at end-diastole and end-systole. These extend segmentation into functional analysis.

Loss Function

The overall loss combines binary cross-entropy, Dice loss, and PaFW regularization:

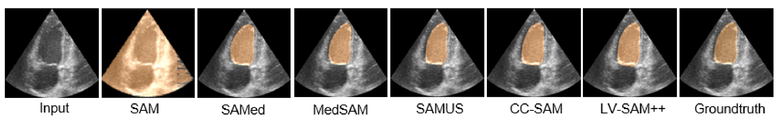
$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} + \lambda_{\text{pfw}}\mathcal{L}_{\text{pfw}}, \tag{2}$$

with λ_{pfw} set to 0.1. Only CNN, adapters, and PaFW are trained, while the ViT backbone and language encoders are frozen for efficiency.

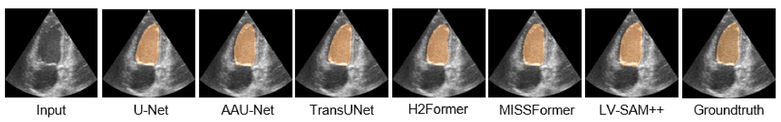
Results

Table 1. Quantitative comparison of LV-SAM++ with foundation and task-specific models on the CAMUS dataset. Bold indicates the best results.

Method	Structural Metrics		Functional Metrics	
	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
Foundation models				
SAM	28.18 ± 7.20	196.64 ± 35.11	–	–
MedSAM	87.52 ± 4.90	15.28 ± 2.75	0.0239 ± 0.0031	0.091 ± 0.010
SAMed	87.67 ± 4.80	13.24 ± 2.31	0.0242 ± 0.0034	0.094 ± 0.011
SAMUS	91.13 ± 3.70	11.76 ± 1.95	0.0172 ± 0.0025	0.079 ± 0.009
CC-SAM	92.85 ± 3.20	10.88 ± 1.80	0.0150 ± 0.0021	0.074 ± 0.008
Task-specific models				
U-Net	93.56 ± 2.90	9.90 ± 1.62	0.0167 ± 0.0023	0.082 ± 0.008
SwinUNET	91.88 ± 3.50	13.11 ± 2.12	0.0192 ± 0.0027	0.087 ± 0.009
SETR	92.77 ± 3.10	11.50 ± 1.88	0.0167 ± 0.0024	0.082 ± 0.008
MISSFormer	93.25 ± 2.95	9.94 ± 1.64	0.0163 ± 0.0021	0.084 ± 0.007
TransUNet	93.60 ± 2.85	9.60 ± 1.55	0.0155 ± 0.0020	0.080 ± 0.007
AAU-Net	93.59 ± 2.82	9.20 ± 1.48	0.0160 ± 0.0021	0.078 ± 0.006
H2Former	93.44 ± 2.88	9.60 ± 1.51	0.0153 ± 0.0020	0.077 ± 0.006
LV-SAM++ (Ours)	**93.97 ± 2.65**	**9.14 ± 1.42**	**0.0142 ± 0.0018**	**0.071 ± 0.006**



(a) **Foundation models.** LV-SAM++ generates more accurate LV boundaries on CAMUS.



(b) **Task-specific models.** LV-SAM++ shows more continuous contours and fewer errors.

Figure 3. Qualitative comparison.

Functional Evaluation

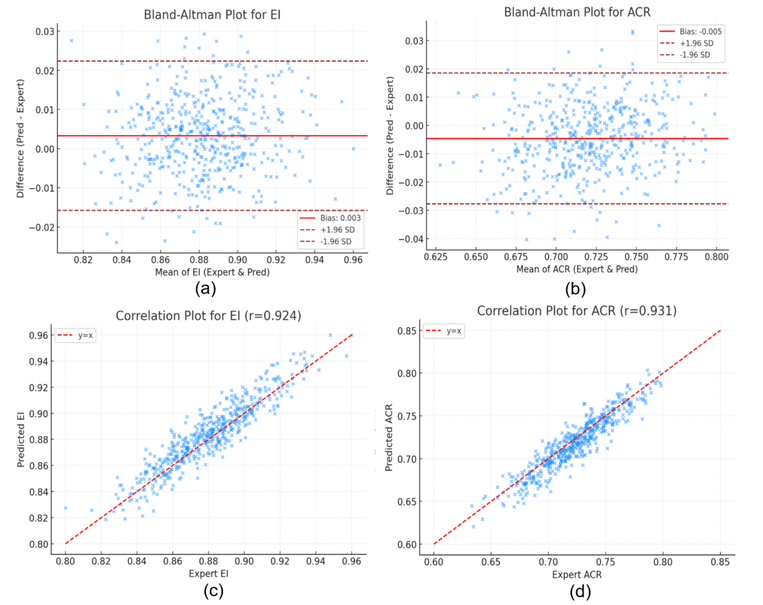


Figure 4. Functional evaluation on CAMUS: (a,b) Bland–Altman plots for EI and ACR show small biases and narrow limits of agreement. (c,d) Correlation plots demonstrate strong agreement with expert measurements.

Ablation Experiment

The largest drop occurs without LFE, confirming its importance for delineating blurry boundaries. Excluding SPL or HSPF mainly affects functional metrics, showing that semantic prompts improve clinical parameter estimation. Overall, all modules complement each other, jointly enhancing segmentation accuracy and the accuracy of functional index estimation.

Table 2. Ablation results of LV-SAM++ on CAMUS. Bold indicates the best.

Configuration	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
w/o PaFW	93.21	9.52	0.0150	0.078
w/o HSPF (SPL)	93.08	9.63	0.0154	0.079
w/o SPL branch	92.84	9.88	0.0162	0.081
w/o LFE	92.60	10.02	0.0167	0.083
Full LV-SAM++	**93.97**	**9.14**	**0.0142**	**0.071**

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